

Publishing while Female — Main Results Summary

April 2, 2026

Contents

1	Figures	2
2	Table 1 LLM — LLM tonal characteristics by gender	8
3	Table 3 — Gender differences in readability, article-level (FemRatio)	9
4	Table 3 LLM — Gender differences in LLM tonal composites, article-level (FemRatio)	11
5	Table 3 LLM Individual — Individual LLM tonal criteria, female ratio (G1–G3)	13
6	Table 3 LLM Individual — Individual LLM tonal criteria, female ratio (G4–G5 and Readability)	15
7	Table 4	17
8	Table 5 — Impact of peer review on gender readability gap (FemRatio)	19
9	Table 5 LLM — Impact of peer review on gender LLM tonal composite gap (FemRatio)	21
10	Table 5 LLM Individual — Impact of peer review on individual LLM tonal criteria, female ratio (G1–G3)	23
11	Table 5 LLM Individual — Impact of peer review on individual LLM tonal criteria, female ratio (G4–G5 and Readability)	25
12	Table 8 — Gender gap in readability at increasing t (FemRatio)	27
13	Table 8 LLM — Gender gap in LLM tonal composite scores at increasing t (FemRatio)	28
14	Table 9 — D_{ik} (base)	29
15	Table 9 LLM — D_{ik} (LLM tonal composites, base)	30
16	Table 10	31

1 Figures

1.1 Figure 1

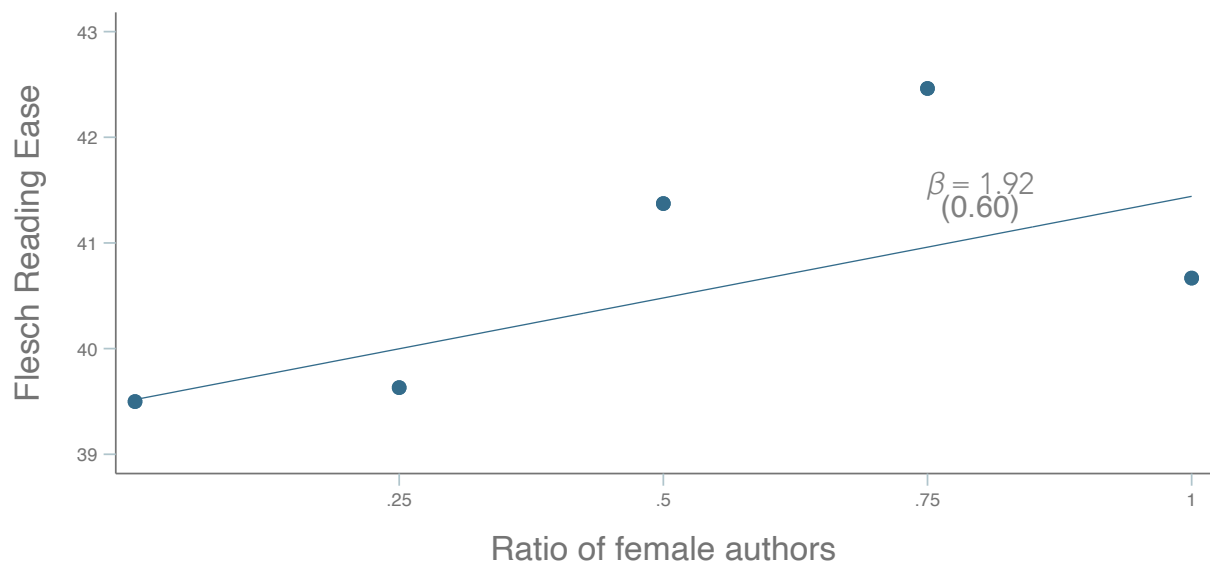
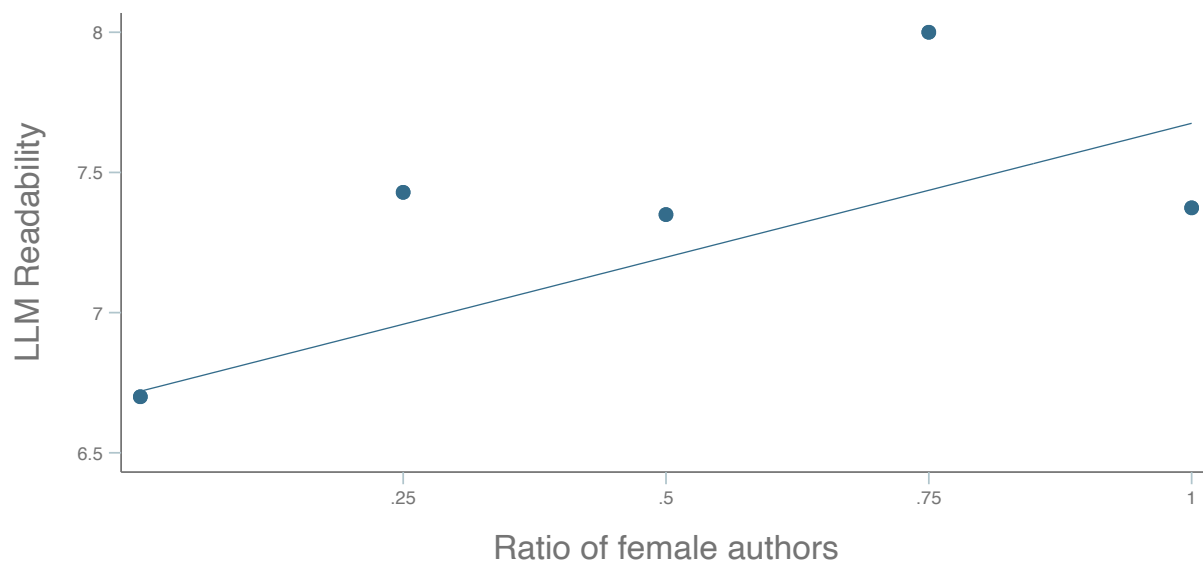


Figure 1: Figure 1

1.2 Figure 1 LLM



$R = 0.96$

Figure 2: Figure 1 LLM

1.3 Figure 4

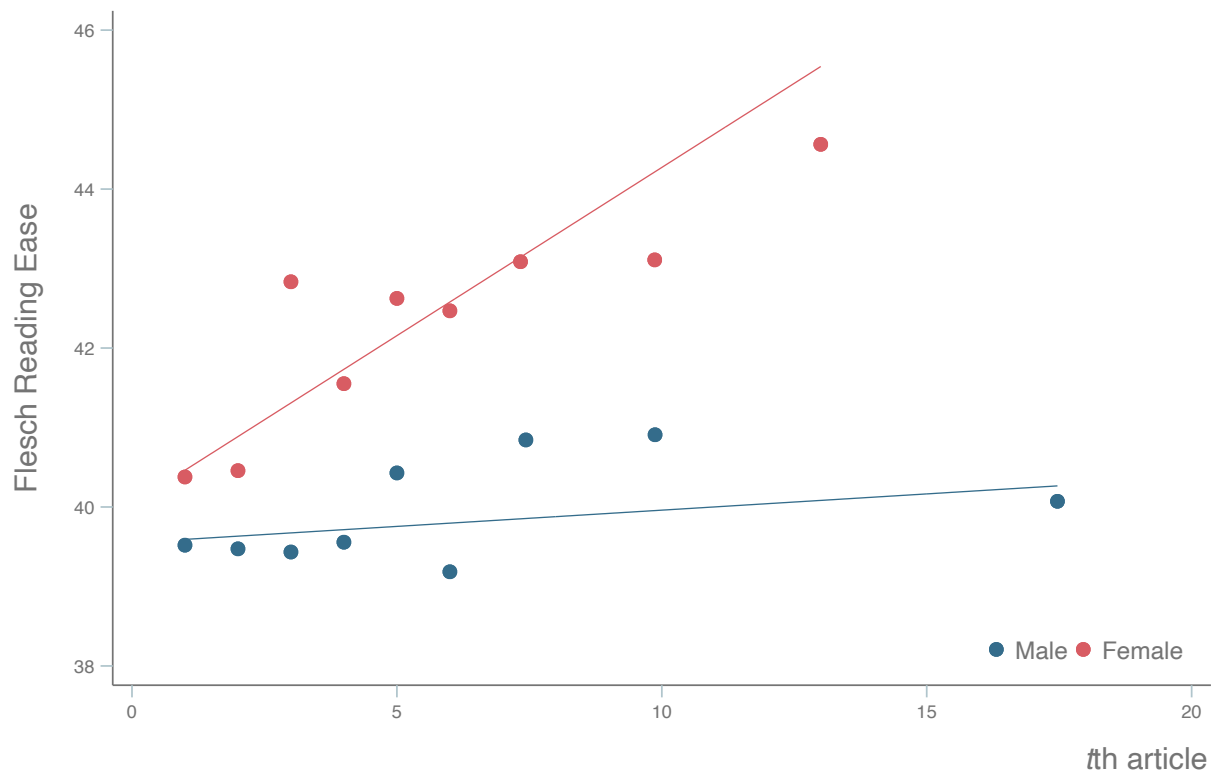


Figure 3: Figure 4

1.4 Figure 4 LLM

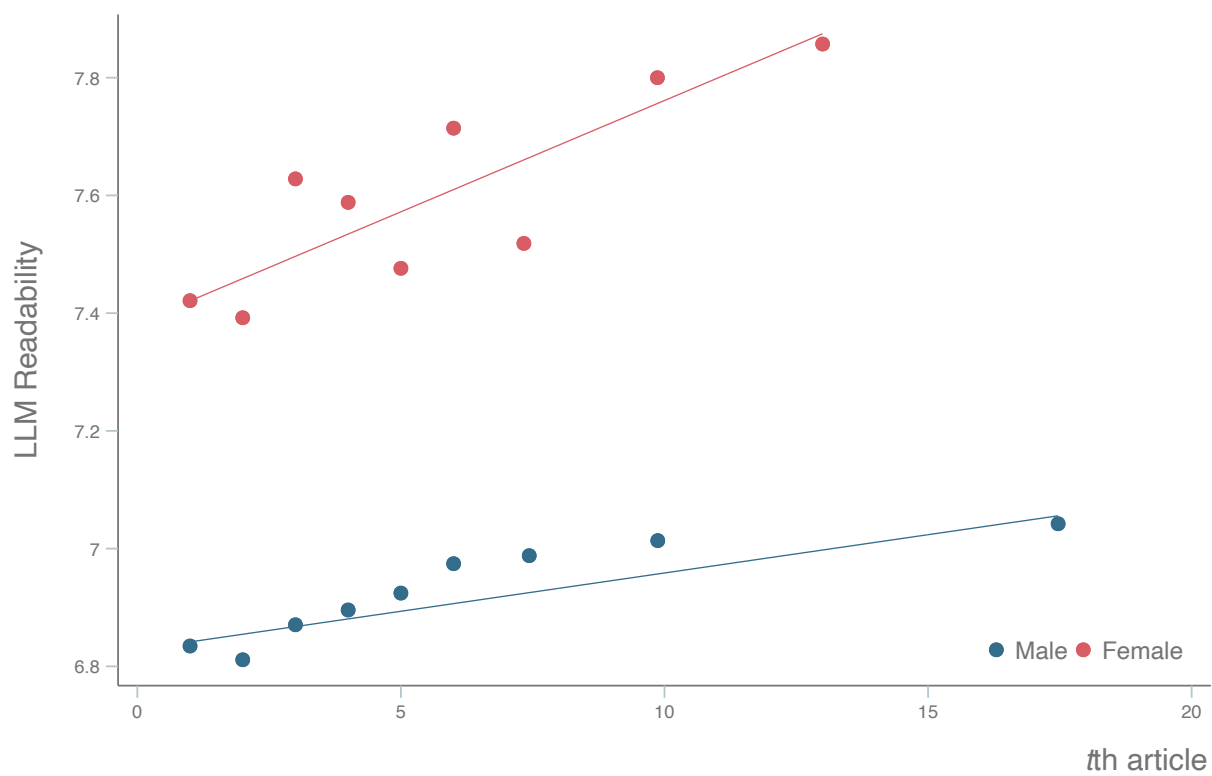


Figure 4: Figure 4 LLM

1.5 Figure 6

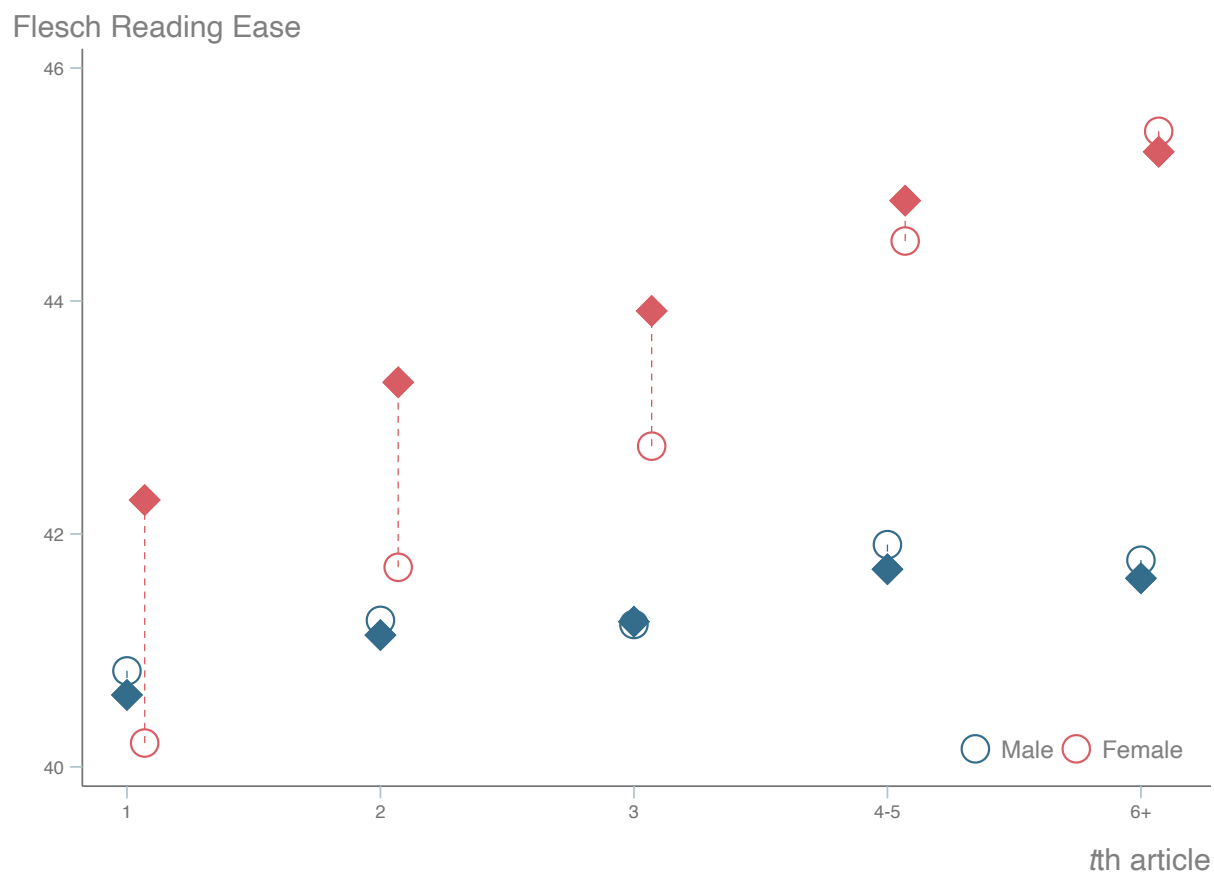


Figure 5: Figure 6

1.6 Figure 6 LLM

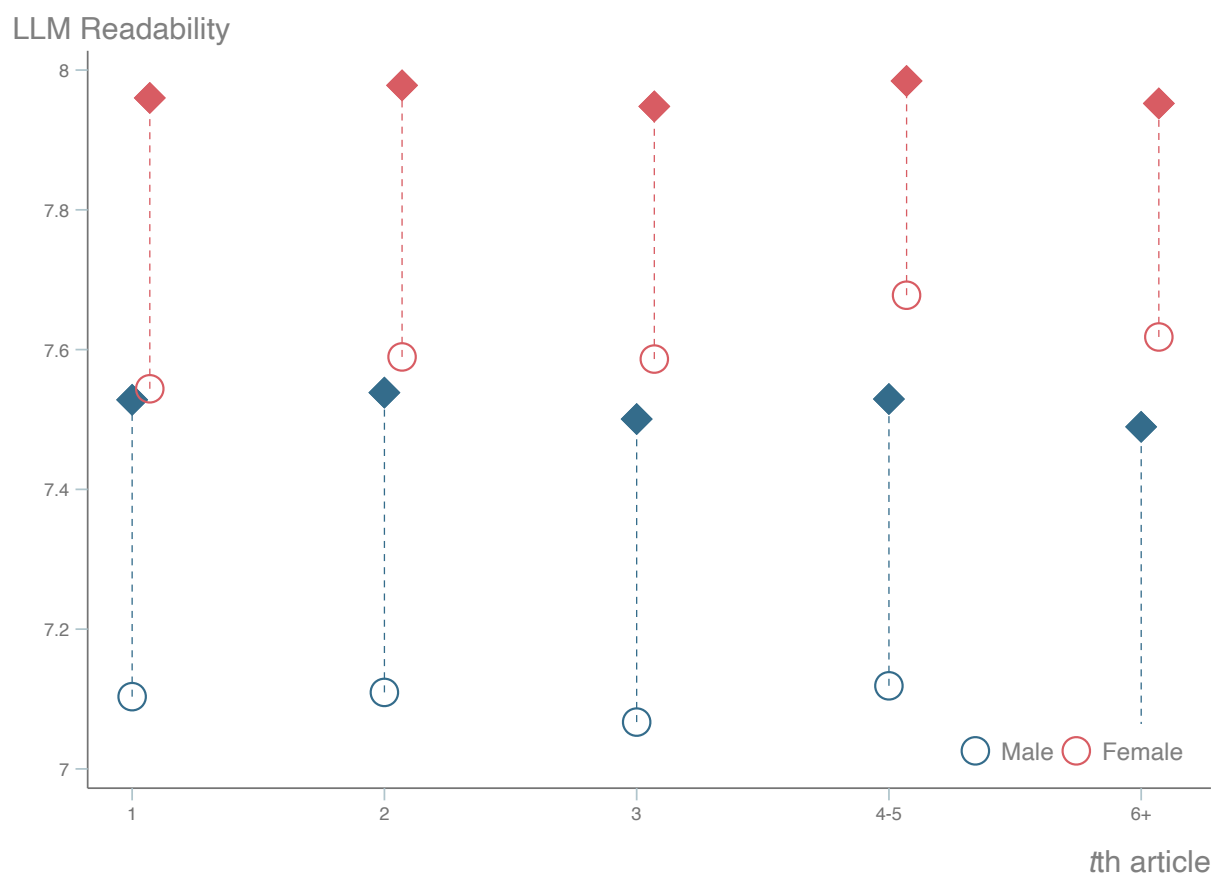


Figure 6: Figure 6 LLM

2 Table 1 LLM — LLM tonal characteristics by gender

Table 1: LLM tonal characteristics by gender

	Men	Women	Difference
LLM Readability	6.73 (0.02)	7.39 (0.05)	0.67 * ** (0.05)
Modal Verb Strength	6.17 (0.03)	6.10 (0.10)	−0.07 (0.11)
Hedging Frequency & Type	8.35 (0.02)	8.34 (0.07)	−0.02 (0.07)
Qualifier Density	8.24 (0.02)	8.30 (0.06)	0.06 (0.07)
Acknowledgement of Limitations	8.05 (0.03)	8.46 (0.09)	0.41 * ** (0.09)
Caution-Signaling Connectors	8.61 (0.02)	8.54 (0.08)	−0.08 (0.08)
Assertiveness & Voice	8.20 (0.02)	8.27 (0.05)	0.07 (0.06)
Active/Passive Voice Ratio	7.22 (0.03)	8.24 (0.09)	1.02 * ** (0.10)
Sentence Length & Directness	5.97 (0.02)	6.63 (0.07)	0.66 * ** (0.07)
Imperative-Form Occurrence	1.02 (0.00)	1.00 (0.01)	−0.02 * * (0.01)
Pronoun Commitment	4.19 (0.04)	5.40 (0.12)	1.21 * ** (0.13)
Novelty-Claim Strength	2.27 (0.02)	2.11 (0.07)	−0.15 * * (0.08)
Jargon/Technicality Density	4.02 (0.02)	4.69 (0.06)	0.67 * ** (0.06)
Emotional Valence	1.12 (0.00)	1.11 (0.01)	−0.01 (0.01)
Evidence & Citation Usage	1.70 (0.02)	2.24 (0.05)	0.54 * ** (0.05)
Practical/Impact Orientation	2.37 (0.02)	2.61 (0.06)	0.23 * ** (0.06)
No. observations	8.268849		9.117

Notes. Figures are means of LLM-scored tonal characteristics by sex. Male-authored papers are defined as having a ratio of female authors below 50 percent; female-authored papers are those with a ratio of female authors at or above 50 percent. Last column subtracts male means from female means. Standard errors in parentheses. ***, ** and * difference statistically significant at 1%, 5% and 10%, respectively.

3 Table 3 — Gender differences in readability, article-level (FemRatio)

Table 2: Gender differences in readability, article-level analysis

	1950–2015					1990–2015			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Flesch	1.25 ** (0.53)	1.23 ** (0.52)	1.23 ** (0.53)	1.26 ** (0.54)	1.49 ** (0.56)	1.31 ** (0.59)	1.36 ** (0.54)	1.29 ** (0.60)	1.94 ** (0.77)
Flesch-Kincaid	0.23* (0.13)	0.22* (0.13)	0.23* (0.13)	0.24* (0.13)	0.26* (0.14)	0.34 ** (0.15)	0.37 ** (0.14)	0.34 ** (0.15)	0.46 ** (0.17)
Gunning Fog	0.36 ** (0.15)	0.36 ** (0.15)	0.37 ** (0.15)	0.40 ** (0.15)	0.42 ** (0.16)	0.51 ** (0.18)	0.50 ** (0.15)	0.51 ** (0.18)	0.60 ** (0.19)
SMOG	0.25 ** (0.10)	0.25 ** (0.10)	0.26 ** (0.10)	0.27 ** (0.11)	0.29 ** (0.11)	0.32 ** (0.12)	0.31 ** (0.11)	0.33 ** (0.13)	0.40 ** (0.14)
Dale-Chall	0.08 (0.05)	0.08 (0.05)	0.08 (0.05)	0.08 (0.05)	0.09 (0.05)	0.11* (0.06)	0.11* (0.06)	0.11* (0.06)	0.17 ** (0.07)
LLM Readability	0.36 ** (0.07)	0.35 ** (0.07)	0.35 ** (0.07)	0.40 ** (0.08)	0.32 ** (0.08)	0.39 ** (0.08)	0.31 ** (0.06)	0.38 ** (0.07)	0.27 ** (0.06)
No. obs.	9.117	9.117	9.117	9.117	9.117	5.211	5.211	5.211	5.774
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓	✓							
Year		✓							
Journal×Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality					✓ ¹	✓ ¹	✓ ¹	✓ ¹	✓ ¹
Native speaker					✓	✓	✓	✓	✓
<i>JEL</i> (primary)							✓		
Theory/empirical									
<i>JEL</i> (tertiary)								✓	✓

Notes. Figures represent coefficients on female ratio from 45 separate OLS regressions of readability scores on the ratio of female authors (papers with fewer than 50 percent female authors are classified as male, see the text). (6)–(9) are estimated on the sample of papers published on or after 1990 with a primary *JEL* code; (9) includes 563 articles from *AER P&P* (see the paper). Quality controls denoted by ✓¹ include citation count (asinh), max. T fixed effects (author prominence) and max. t (author seniority). Standard errors clustered on editor in parentheses. ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

4 Table 3 LLM — Gender differences in LLM tonal composites, article-level (FemRatio)

Table 3: Gender differences in LLM tonal composites, article-level analysis

	1950–2015					1990–2015			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
LLM G1: Creativity & Hedging	−0.05	−0.04	−0.06	−0.04	−0.05	−0.04	−0.02	−0.04	−0.10
LLM G2: Assertiveness & Voice	(0.08) −0.01	(0.08) −0.01	(0.08) −0.01	(0.08) 0.05	(0.08) 0.05	(0.09) 0.06	(0.09) 0.07	(0.09) 0.07	(0.09) 0.08
LLM G3: Structural Directness	(0.07) 0.13 **	(0.07) 0.12 **	(0.07) 0.12 **	(0.07) 0.14 ***	(0.07) 0.12 **	(0.07) 0.18 ***	(0.08) 0.17 ***	(0.07) 0.18 ***	(0.08) 0.15 ***
LLM G4: Authorial Stance & Novelty	(0.05) 0.02	(0.05) 0.02	(0.05) 0.02	(0.05) 0.15 ***	(0.05) 0.12*	(0.05) 0.17 **	(0.05) 0.14 **	(0.05) 0.17 **	(0.05) 0.17 ***
LLM G5: Support & Impact	(0.05) 0.25 ***	(0.05) 0.25 ***	(0.05) 0.26 ***	(0.06) 0.27 ***	(0.06) 0.20 **	(0.07) 0.18*	(0.06) 0.07	(0.07) 0.16*	(0.06) 0.00
	(0.07)	(0.08)	(0.08)	(0.08)	(0.08)	(0.10)	(0.09)	(0.09)	(0.08)
No. obs.	9.117	9.117	9.117	9.117	9.117	5.211	5.211	5.211	5.774
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓	✓							
Year	✓	✓							
Journal × Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality					✓ ¹	✓ ¹	✓ ¹	✓ ¹	✓ ¹
Native speaker					✓	✓	✓	✓	✓
<i>JEL</i> (primary)							✓	✓	
Theory/empirical								✓	
<i>JEL</i> (tertiary)									✓

Notes. Figures represent coefficients on female ratio from 45 separate OLS regressions of LLM tonal composite scores on the ratio of female authors. Each row corresponds to a separate LLM composite group. ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

**5 Table 3 LLM Individual — Individual LLM tonal
criteria, female ratio (G1–G3)**

Table 4: Gender differences in individual LLM tonal criteria, article-level analysis (G1–G3)

	1950–2015				1990–2015				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
G1: Creativity & Hedging									
Modal Verb Strength	−0.32* (0.18)	−0.33* (0.18)	−0.34* (0.19)	−0.32* (0.19)	−0.34* (0.19)	−0.29 (0.22)	−0.26 (0.21)	−0.30 (0.22)	−0.31 (0.21)
Hedging Frequency & Type	−0.21* (0.11)	−0.21* (0.11)	−0.23** (0.11)	−0.20* (0.11)	−0.17 (0.11)	−0.19 (0.13)	−0.14 (0.13)	−0.19 (0.13)	−0.19 (0.12)
Qualifier Density	−0.02 (0.10)	−0.01 (0.10)	−0.03 (0.10)	0.00 (0.09)	0.01 (0.10)	0.06 (0.10)	0.07 (0.10)	0.06 (0.10)	−0.06 (0.12)
Acknowledgement of Limitations	0.23* (0.12)	0.24** (0.12)	0.23* (0.12)	0.24** (0.11)	0.15 (0.12)	0.18 (0.13)	0.14 (0.13)	0.17 (0.13)	0.10 (0.14)
Caution-Signaling Connectors	0.06 (0.13)	0.08 (0.13)	0.09 (0.13)	0.08 (0.14)	0.08 (0.14)	0.04 (0.17)	0.07 (0.18)	0.04 (0.17)	−0.03 (0.18)
G2: Assertiveness & Voice									
Assertiveness & Voice	−0.08 (0.08)	−0.08 (0.08)	−0.10 (0.08)	−0.06 (0.07)	−0.05 (0.08)	−0.07 (0.09)	−0.04 (0.09)	−0.07 (0.09)	−0.07 (0.09)
Active/Passive Voice Ratio	0.07 (0.10)	0.06 (0.10)	0.07 (0.10)	0.16 (0.10)	0.15 (0.10)	0.18* (0.11)	0.19* (0.11)	0.20* (0.11)	0.23** (0.11)
G3: Structural Directness									
Sentence Length & Directness	0.27** (0.10)	0.27** (0.11)	0.27** (0.11)	0.31** (0.10)	0.25** (0.10)	0.37*** (0.10)	0.36*** (0.10)	0.37*** (0.10)	0.31*** (0.10)
Imperative-Form Occurrence	−0.02*** (0.01)	−0.02*** (0.01)	−0.02*** (0.01)	−0.02*** (0.01)	−0.02*** (0.01)	−0.01** (0.01)	−0.01** (0.01)	−0.01** (0.01)	−0.01* (0.00)
No. obs.	9.117	9.117	9.117	9.117	9.117	5.211	5.211	5.211	5.774
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓	✓							
Year		✓							
Journal×Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality					✓ ¹	✓ ¹	✓ ¹	✓ ¹	✓ ¹
Native speaker					✓	✓	✓	✓	✓
<i>JEL</i> (primary)									
Theory/empirical									
<i>JEL</i> (tertiary)								✓	✓

Notes. Figures represent coefficients on female ratio from 81 separate OLS regressions of individual L1M tonal criteria on the ratio of female authors. Each row is a separate criterion. Criteria belong to groups G1–G3.

Notes. Figures represent coefficients on female ratio from 81 separate OLS regressions of individual LLM tonal criteria on the ratio of female authors. Each row is a separate criterion. Criteria belong to groups G1 (Creativity & Hedging), G2 (Assertiveness & Voice), and G3 (Structural Directness). ***, **, * and * statistically significant at 1%, 5% and 10%, respectively.

**6 Table 3 LLM Individual — Individual LLM tonal
criteria, female ratio (G4–G5 and Readability)**

Table 5: Gender differences in individual LLM tonal criteria, article-level analysis (G4–G5 and Readability)

	1950–2015			1990–2015					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
G4: Authorial Stance & Novelty									
Pronoun Commitment	−0.09 (0.18)	−0.10 (0.18)	−0.10 (0.18)	0.39 ** (0.19)	0.32 (0.20)	0.53 ** (0.24)	0.54 ** (0.23)	0.53 ** (0.24)	0.61 *** (0.20)
Novelty-Claim Strength	−0.32 *** (0.10)	−0.32 *** (0.11)	−0.32 *** (0.11)	−0.31 *** (0.11)	−0.30 *** (0.11)	−0.36 *** (0.14)	−0.30 *** (0.14)	−0.36 *** (0.14)	−0.30 *** (0.13)
Jargon/Technicality Density [†]	0.52 ***	0.52 ***	0.53 ***	0.55 ***	0.48 ***	0.54 ***	0.37 ***	0.52 ***	0.41 ***
Emotional Valence	(0.11)	(0.11)	(0.11)	(0.12)	(0.12)	(0.15)	(0.11)	(0.14)	(0.12)
	−0.02 (0.02)	−0.02 (0.02)	−0.03 (0.02)	−0.03 (0.02)	−0.03 (0.02)	−0.03 (0.02)	−0.03 (0.02)	−0.03 (0.02)	−0.03 (0.02)
G5: Support & Impact									
Evidence & Citation Usage	0.44 *** (0.09)	0.44 *** (0.09)	0.44 *** (0.09)	0.46 *** (0.09)	0.36 *** (0.09)	0.37 *** (0.11)	0.26 *** (0.10)	0.33 *** (0.11)	0.16 (0.10)
Practical/Impact Orientation	0.06 (0.09)	0.06 (0.09)	0.07 (0.10)	0.09 (0.09)	0.05 (0.10)	−0.01 (0.11)	−0.11 (0.10)	−0.01 (0.11)	−0.15 (0.09)
Standalone									
Readability	0.36 *** (0.07)	0.35 *** (0.07)	0.35 *** (0.07)	0.40 *** (0.08)	0.32 *** (0.08)	0.39 *** (0.08)	0.31 *** (0.06)	0.38 *** (0.07)	0.27 *** (0.06)
No. obs.	9.117	9.117	9.117	9.117	9.117	5.211	5.211	5.211	5.774
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓	✓							
Year		✓							
Journal×Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality					✓ ¹	✓ ¹	✓ ¹	✓ ¹	✓ ¹
Native speaker					✓	✓	✓	✓	✓
<i>JEL</i> (primary)									
Theory/empirical									
<i>JEL</i> (tertiary)								✓	✓

Notes. Figures represent coefficients on female ratio from 63 separate OLS regressions of individual LLM tonal criteria on the ratio of female authors. Each row is a separate criterion. Criteria belong to groups G4 (Authorial Stance & Novelty), G5 (Support & Impact), and the standalone Readability criterion. [†] Jargon/Technicality Density is scale-flipped (11 – raw score) so that higher values indicate more accessible language. See companion table for G1–G3. **, *, ***, and * statistically significant at 1%, 5% and 10%, respectively.

7 Table 4

Table 6: Textual characteristics, published papers vs. drafts

	Men			Women			Diff.-in diff.
	Working paper	Published article	Difference	Working paper	Published article	Difference	
No. sentences	6.48 (0.06)	5.10 (0.04)	-1.376 * ** (0.054)	6.76 (0.15)	5.06 (0.08)	-1.702 * ** (0.139)	-0.327 * * (0.149)
No. characters	862.64 (7.19)	649.74 (4.67)	-212.899 * ** (7.162)	905.83 (18.54)	636.18 (10.35)	-269.654 * ** (18.439)	-56.660 * ** (19.632)
No. words	155.63 (1.32)	115.69 (0.85)	-39.941 * ** (1.324)	164.05 (3.42)	113.67 (1.92)	-50.386 * ** (3.424)	-10.403 * ** (3.636)
No. syllables	257.06 (2.15)	193.33 (1.40)	-63.730 * ** (2.136)	268.45 (5.54)	187.78 (3.09)	-80.669 * ** (5.500)	-16.894 * ** (5.852)
No. polysyllabic words	28.40 (0.28)	21.80 (0.18)	-6.603 * ** (0.245)	28.89 (0.71)	20.62 (0.41)	-8.265 * ** (0.625)	-1.648 * * (0.669)
No. difficult words	58.45 (0.51)	44.64 (0.33)	-13.815 * ** (0.482)	60.16 (1.30)	42.42 (0.74)	-17.746 * ** (1.200)	-3.923 * ** (1.316)
Flesch	41.40 (0.26)	41.15 (0.18)	-0.248 (0.184)	42.51 (0.66)	43.12 (0.42)	0.611 (0.451)	0.828* (0.500)
Flesch-Kincaid	-13.63 (0.06)	-13.38 (0.05)	0.252 * ** (0.049)	-13.53 (0.15)	-13.00 (0.11)	0.531 * ** (0.122)	0.273 * * (0.134)
Gunning Fog	-17.28 (0.07)	-17.03 (0.05)	0.254 * ** (0.055)	-17.12 (0.18)	-16.58 (0.13)	0.545 * ** (0.140)	0.281* (0.149)
SMOG	-15.14 (0.05)	-15.00 (0.03)	0.145 * ** (0.035)	-15.02 (0.13)	-14.70 (0.09)	0.327 * ** (0.094)	0.179* (0.097)
Dale-Chall	-10.84 (0.02)	-10.93 (0.02)	-0.090 * ** (0.015)	-10.70 (0.06)	-10.71 (0.04)	-0.004 (0.037)	0.086 * * (0.042)
LLM Readability	7.03 (0.03)	7.46 (0.02)	0.431 * ** (0.024)	7.46 (0.07)	7.87 (0.05)	0.408 * ** (0.056)	-0.033 (0.066)
LLM G1: Creativity & Hedging	7.61 (0.04)	7.96 (0.03)	0.349 * ** (0.034)	7.53 (0.09)	7.88 (0.07)	0.357 * ** (0.084)	0.002 (0.092)
LLM G2: Assertiveness & Voice	8.23 (0.03)	8.42 (0.02)	0.191 * ** (0.026)	8.32 (0.06)	8.49 (0.05)	0.169 * ** (0.062)	-0.023 (0.071)
LLM G3: Structural Directness	3.38 (0.02)	3.86 (0.02)	0.486 * ** (0.023)	3.49 (0.06)	4.03 (0.04)	0.544 * ** (0.059)	0.059 (0.063)
LLM G4: Authorial Stance & Novelty	3.53 (0.03)	3.38 (0.02)	-0.158 * ** (0.020)	3.87 (0.06)	3.68 (0.04)	-0.195 * ** (0.045)	-0.040 (0.053)
LLM G5: Support & Impact	2.83 (0.04)	2.56 (0.02)	-0.273 * ** (0.026)	3.12 (0.09)	2.88 (0.06)	-0.239 * ** (0.070)	0.023 (0.072)
No. observations	1.716	1.716	1.716	272	272	272	1.988

Notes. Figures are means of textual characteristics by sex for NBER working papers and published articles. Male-authored papers are defined as having a ratio of female authors below 50 percent; female-authored papers are those with a ratio of female authors at or above 50 percent. Penultimate columns in each panel subtract working paper figures from published article figures for men (first panel) and women (second panel); difference-in-differences (female less male) shown in the final column. Standard errors in parentheses. ***, ** and * difference statistically significant at 1%, 5% and 10%, respectively.

8 Table 5 — Impact of peer review on gender readability gap (FemRatio)

Table 7: The impact of peer review on the gender readability gap

OLS (score)			OLS (Δ in score)			FGLS (score)		
R_{jw}	Working paper			Published paper			Difference	
	Female ratio	Blind $\times$ fem. ratio	Female ratio	Blind $\times$ fem. ratio	Female ratio	Blind $\times$ fem. ratio	Female ratio	Blind $\times$ fem. ratio
Flesch	0.83 *** (0.02)	-2.15 (3.84)	1.19* (0.70)	-1.75 (2.18)	1.68 (1.05)	-2.35 (3.37)	2.87 *** (1.08)	-4.10* (2.16)
Flesch-Kincaid	0.74 *** (0.04)	-0.92 (0.83)	0.48 *** (0.20)	-0.74 (0.52)	0.18 (0.26)	-0.70 (0.77)	0.66 *** (0.27)	-1.44 *** (0.44)
Gunning Fog	0.76 *** (0.03)	-0.93 (0.80)	0.48 *** (0.22)	-0.76 (0.57)	0.26 (0.28)	-0.71 (0.85)	0.74 *** (0.28)	-1.47 *** (0.53)
SMOG	0.79 *** (0.03)	-0.59 (0.56)	0.27* (0.14)	-0.48 (0.39)	0.20 (0.17)	-0.52 (0.61)	0.47 *** (0.18)	-1.00 *** (0.40)
Dale-Chall	0.85 *** (0.02)	-0.20 (0.20)	0.14 *** (0.06)	-0.20 (0.13)	0.21 *** (0.10)	-0.04 (0.23)	0.35 *** (0.10)	-0.24 (0.20)
LLM Readability	0.50 *** (0.02)	-0.09 (0.16)	-0.08 (0.08)	0.04 (0.17)	0.50 *** (0.09)	-0.27* (0.15)	0.42 *** (0.08)	-0.22 (0.21)
No. observations	1.988	1.988	1.988	1.988	1.988	1.988	1.988	1.988
Editor effects	✓		✓		✓		✓	
Journal $\times$ Year effects	✓		✓		✓		✓	
N_j	✓		✓		✓		✓	
Quality controls	✓ ²		✓ ²		✓ ²		✓ ²	
Native speaker	✓		✓		✓		✓	
Theory/emp. effects	✓		✓		✓		✓	

Notes. Panel one displays coefficients from estimating equation (2) directly via OLS; standard errors clustered by editor in parentheses. Panel two displays coefficients from OLS estimation of equation (3); standard errors clustered by year in parentheses. Panel three displays coefficients from FGLS estimation of equation (6) and equation (7); standard errors clustered by year and robust to cross-model correlation in parentheses. Quality controls denoted by $\checkmark^2$ include citation count (asinh), max. t (author prominence) and max. T (author seniority). The variable "female ratio" defines papers with a strict minority of female authors as male-authored; for papers with 50 percent or more female authors, it is the ratio of female authors on a paper (see the text for more details). ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

**9 Table 5 LLM — Impact of peer review on gender
LLM tonal composite gap (FemRatio)**

Table 8: The impact of peer review on the gender LLM tonal composite gap

	OLS (score)			OLS (Δ in score)			FGLS (score)		
	R_{jw}	Female	Blind $\times$ female	Female	Blind $\times$ female	Female	Working paper	Published paper	Difference
LLM G1: Creativity & Hedging	0.48 *** (0.02)	-0.19 (0.12)	0.21 (0.21)	-0.14 (0.13)	-0.05 (0.25)	-0.10 (0.13)	0.51 (0.33)	-0.24 ** (0.12)	-0.14 -0.05 (0.13) (0.24)
LLM G2: Assertiveness & Voice	0.52 *** (0.02)	0.05 (0.09)	-0.41 ** (0.16)	0.03 (0.15)	-0.47 (0.38)	0.04 (0.12)	0.12 (0.19)	0.07 (0.12)	0.03 -0.47 (0.15) (0.37)
LLM G3: Structural Directness	0.38 *** (0.03)	0.17 ** (0.08)	-0.40 *** (0.14)	0.10 (0.10)	-0.15 (0.15)	0.11 (0.08)	-0.41 ** (0.16)	0.21 ** (0.09)	0.10 -0.15 (0.10) (0.14)
LLM G4: Authorial Stance & Novelty	0.65 *** (0.03)	0.06 (0.09)	-0.06 (0.16)	-0.07 (0.07)	-0.01 (0.17)	0.37 ** (0.14)	-0.13 (0.20)	0.30 ** (0.14)	-0.07 -0.01 (0.07) (0.17)
LLM G5: Support & Impact	0.67 *** (0.02)	0.05 (0.09)	0.11 (0.18)	-0.02 (0.14)	0.01 (0.35)	0.21 (0.16)	0.32 (0.31)	0.19 (0.20)	-0.02 0.01 (0.14) (0.34)
No. observations		1.988		1.988		1.988		1.988	1.988
Editor effects	✓	✓	✓	✓	✓	✓	✓	✓	
Journal $\times$ Year effects	✓	✓	✓	✓	✓	✓	✓	✓	
N_j	✓	✓	✓	✓	✓	✓	✓	✓	
Quality controls	✓ ²	✓ ²	✓	✓ ²	✓	✓ ²	✓	✓ ²	
Native speaker	✓	✓	✓	✓	✓	✓	✓	✓	
Theory/emp. effects	✓	✓	✓	✓	✓	✓	✓	✓	

Notes. Panel one displays coefficients from estimating directly via OLS; standard errors clustered by editor. Panel two uses the change in LLM tonal composite score as the dependent variable. Panel three uses FGLS. Each row corresponds to a separate LLM composite group. ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

10 Table 5 LLM Individual — Impact of peer review on individual LLM tonal criteria, female ratio (G1–G3)

Table 9: Gender differences in individual LLM tonal criteria, impact of peer review (G1–G3)

	OLS (score)			OLS (Δ in score)			FGLS (score)		
	R_{jW}	Female	Blind $\times$ female	Female	Blind $\times$ female	Female	Working paper	Published paper	Difference
G1: Creativity & Hedging									
Modal Verb Strength	0.34 *** (0.02)	-0.59* (0.36)	1.00* (0.53)	-0.50* (0.30)	0.17 (0.74)	-0.14 (0.30)	1.26* (0.74)	-0.64* (0.35)	0.17 (0.72)
Hedging Frequency & Type	0.43 *** (0.03)	-0.29* (0.17)	0.14 (0.42)	-0.12 (0.18)	-0.78* (0.42)	-0.30 (0.19)	1.62 *** (0.48)	-0.42 ** (0.18)	-0.78* (0.41)
Qualifier Density	0.42 *** (0.02)	0.03 (0.13)	-0.04 (0.21)	0.06 (0.18)	-0.24 (0.71)	-0.05 (0.16)	0.34 (0.58)	0.01 (0.15)	-0.24 (0.70)
Acknowledgement of Limitations	0.45 *** (0.03)	0.09 (0.23)	-0.02 (0.25)	-0.02 (0.27)	0.02 (0.53)	0.21 (0.22)	-0.08 (0.60)	0.19 (0.20)	0.02 (0.52)
Caution-Signaling Connectors	0.46 *** (0.03)	-0.23 (0.31)	0.25 (0.43)	-0.11 (0.30)	0.57 (0.72)	-0.22 (0.25)	-0.59 (0.87)	-0.33 (0.29)	0.57 (0.71)
G2: Assertiveness & Voice									
Assertiveness & Voice	0.48 *** (0.02)	-0.14 (0.11)	-0.50 ** (0.25)	-0.13 (0.14)	-0.89 *** (0.29)	-0.01 (0.13)	0.74 ** (0.32)	-0.14 (0.15)	-0.89 *** (0.29)
Active/Passive Voice Ratio	0.56 *** (0.03)	0.24 ** (0.12)	-0.26 (0.32)	0.20 (0.21)	-0.05 (0.59)	0.09 (0.17)	-0.49 (0.48)	0.29* (0.17)	-0.05 (0.58)
G3: Structural Directness									
Sentence Length & Directness	0.38 *** (0.03)	0.34 ** (0.16)	-0.81 *** (0.29)	0.20 (0.20)	-0.33 (0.29)	0.23 (0.16)	-0.78 ** (0.32)	0.43 ** (0.17)	-0.33 (0.29)
No. observations		1.988		1.988		1.988		1.988	
Editor effects		✓		✓		✓		✓	
Journal $\times$ Year effects		✓		✓		✓		✓	
N_j		✓		✓		✓		✓	
Quality controls		✓ ²		✓ ²		✓ ²		✓ ²	
Native speaker		✓		✓		✓		✓	
Theory/emp. effects		✓		✓		✓		✓	

Notes. Panel one displays coefficients from estimating directly via OLS; standard errors clustered by editor. Panel two uses the change in individual LLM tonal criterion score as the dependent variable. Panel three uses FGLS. Each row corresponds to a separate individual LLM criterion. ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

11 Table 5 LLM Individual — Impact of peer review on individual LLM tonal criteria, female ratio (G4–G5 and Readability)

Table 10: Gender differences in individual LLM tonal criteria, impact of peer review (G4–G5 and Readability)

		OLS (score)		OLS (Δ in score)		FGLS (score)					
						Working paper		Published paper		Difference	
						Blind $\times$ female		Blind $\times$ female		Blind $\times$ Female female	
R_{jw}		Female		Female		Female		Female		Female female	
G4: Authorial Stance & Novelty											
Pronoun Commitment		0.70 *** (0.04)	0.18 (0.25)	-0.48 (0.48)	-0.06 (0.25)	-0.33 (0.73)	0.81* (0.47)	-0.51 (0.86)	0.75 (0.48)	-0.84 (0.73)	-0.33 (0.72)
Novelty-Claim Strength		0.58 *** (0.03)	-0.19 (0.15)	-0.12 (0.20)	-0.01 (0.12)	-0.15 (0.32)	-0.42 *** (0.21)	0.07 (0.46)	-0.43 *** (0.22)	-0.07 (0.42)	-0.15 (0.32)
Jargon/Technicality Density [†]		0.74 *** (0.03)	0.11 (0.14)	0.39 *** (0.15)	-0.17 (0.16)	0.38 (0.32)	1.06 *** (0.12)	0.02 (0.34)	0.89 *** (0.14)	0.40 (0.32)	0.38 (0.15)
Emotional Valence		0.49 *** (0.04)	-0.02 (0.04)	-0.01 (0.08)	-0.03 (0.03)	0.05 (0.11)	0.02 (0.05)	-0.11 (0.09)	-0.01 (0.05)	-0.06 (0.07)	0.05 (0.11)
G5: Support & Impact											
Evidence & Citation Usage		0.64 *** (0.03)	0.20 (0.15)	-0.33 (0.30)	0.07 (0.20)	-0.53 (0.39)	0.36* (0.19)	0.56 (0.44)	0.44* (0.23)	0.03 (0.41)	-0.53 (0.38)
Practical/Impact Orientation		0.67 *** (0.03)	-0.10 (0.11)	0.57 *** (0.15)	-0.12 (0.17)	0.54 (0.42)	0.06 (0.16)	0.08 (0.45)	-0.06 (0.20)	0.62 (0.38)	0.54 (0.41)
Standalone											
Readability		0.50 *** (0.02)	0.17* (0.09)	-0.09 (0.16)	-0.08 (0.08)	0.04 (0.17)	0.50 *** (0.09)	-0.27* (0.15)	0.42 *** (0.08)	-0.22 (0.21)	0.04 (0.17)
No. observations		1.988		1.988		1.988		1.988		1.988	
Editor effects		✓			✓		✓		✓		
Journal $\times$ Year effects		✓			✓		✓		✓		
N_j		✓			✓		✓		✓		
Quality controls		✓ ²			✓ ²		✓ ²		✓ ²		
Native speaker		✓			✓		✓		✓		
Theory/emp. effects		✓			✓		✓		✓		

Notes. Panel one displays coefficients from estimating directly via OLS; standard errors clustered by editor. Panel two uses the change in individual LLM tonal criterion score as the dependent variable. Panel three uses FGLS. Each row corresponds to a separate individual LLM criterion. ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

12 Table 8 — Gender gap in readability at increasing t (FemRatio)

Table 11: Gender gap in readability at increasing t

	$t = 1$	$t = 2$	$t = 3$	$t = 4-5$	$t \geq 6$	All
Flesch	0.72 (0.75)	1.91* (0.99)	4.64 * ** (1.50)	3.24 (2.31)	2.69 (2.01)	2.24 * ** (0.74)
Flesch-Kincaid	0.07 (0.19)	0.23 (0.25)	1.04 * ** (0.25)	0.72 (0.44)	0.46 (0.40)	0.29* (0.16)
Gunning Fog	0.21 (0.21)	0.52* (0.28)	1.29 * ** (0.33)	0.99 * * (0.46)	0.62 (0.41)	0.53 * ** (0.19)
SMOG	0.15 (0.14)	0.40 * * (0.19)	0.81 * ** (0.24)	0.73* (0.38)	0.42 (0.29)	0.42 * ** (0.13)
Dale-Chall	0.04 (0.07)	0.05 (0.10)	0.23* (0.13)	0.37* (0.20)	0.42* (0.23)	0.15 * * (0.07)
LLM Readability	0.35 * ** (0.08)	0.29 * * (0.14)	0.64 * ** (0.18)	0.29 (0.18)	0.21 (0.24)	0.31 * ** (0.08)
No. observations	6.874	2.827	1.675	1.906	2.773	12.008
Editor effects	✓	✓	✓	✓	✓	✓
Blind review	✓	✓	✓	✓	✓	✓
Journal×Year ef- fects	✓	✓	✓	✓	✓	✓
N_j	✓	✓	✓	✓	✓	✓
Institution effects	✓	✓	✓	✓	✓	✓
Quality controls	✓ ³	✓ ³	✓ ³	✓ ³	✓ ³	✓ ¹
Native speaker	✓	✓	✓	✓	✓	✓

Notes. β_1 from FGLS estimation of the baseline FGLS specification. First column restricts sample to authors' first top-four publication ($t = 1$), second column to their second ($t = 2$), etc. Regressions weighted by $1/N_j$. Standard errors (in parentheses) adjusted for two-way clustering (editor and author) and cross-model correlation. Final column estimates from an unweighted population-averaged regression; error correlations specified by an auto-regressive process of order one and standard errors (in parentheses) adjusted for one-way clustering on author. Quality controls denoted by ✓¹ include citation count (asinh), max. T fixed effects (author prominence) and max. t (author seniority); ✓³ includes citation count (asinh) and max. t , only. The variable "female ratio" defines papers with a strict minority of female authors as male-authored; for papers with 50 percent or more female authors, it is the ratio of female authors on a paper (see the text for more details). ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

13 Table 8 LLM — Gender gap in LLM tonal composite scores at increasing t (FemRatio)

Table 12: Gender gap in LLM tonal composite scores at increasing t

	$t = 1$	$t = 2$	$t = 3$	$t = 4-5$	$t \geq 6$	All
LLM G1: Creativity & Hedging	−0.08 (0.10)	−0.28* (0.15)	0.21 (0.21)	0.04 (0.29)	−0.04 (0.28)	−0.11 (0.09)
LLM G2: Assertiveness & Voice	0.02 (0.09)	−0.06 (0.10)	0.03 (0.19)	−0.25 (0.24)	−0.03 (0.21)	0.01 (0.08)
LLM G3: Structural Directness	0.10 (0.07)	0.02 (0.12)	0.43 * ** (0.13)	0.01 (0.14)	0.04 (0.18)	0.06 (0.05)
LLM G4: Authorial Stance & Novelty	−0.05 (0.06)	−0.00 (0.16)	0.14 (0.16)	0.23 (0.16)	0.14 (0.19)	0.06 (0.06)
LLM G5: Support & Impact	0.24 * * (0.12)	0.11 (0.12)	0.46 * * (0.22)	0.47 (0.30)	−0.00 (0.24)	0.17* (0.10)
No. observations	6.874	2.827	1.675	1.906	2.773	12.008
Editor effects	✓	✓	✓	✓	✓	✓
Blind review	✓	✓	✓	✓	✓	✓
Journal×Year effects	✓	✓	✓	✓	✓	✓
N_j	✓	✓	✓	✓	✓	✓
Institution effects	✓	✓	✓	✓	✓	✓
Quality controls	✓ ³	✓ ³	✓ ³	✓ ³	✓ ³	✓ ¹
Native speaker	✓	✓	✓	✓	✓	✓

Notes. β_1 from FGLS estimation of the baseline FGLS specification. First column restricts sample to authors' first top-four publication ($t = 1$), second column to their second ($t = 2$), etc. Regressions weighted by $1/N_j$. Standard errors (in parentheses) adjusted for two-way clustering (editor and author) and cross-model correlation. Final column estimates from an unweighted population-averaged regression; error correlations specified by an auto-regressive process of order one and standard errors (in parentheses) adjusted for one-way clustering on author. Quality controls denoted by ✓¹ include citation count (asinh), max. T fixed effects (author prominence) and max. t (author seniority); ✓³ includes citation count (asinh) and max. t , only. The variable "female ratio" defines papers with a strict minority of female authors as male-authored; for papers with 50 percent or more female authors, it is the ratio of female authors on a paper (see the text for more details). ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

14 Table 9 — D_{ik} (base)

Table 13: D_{ik} (??)

	Higher standards for women ($D_{ik} > 0$)			Higher standards for men ($D_{ik} < 0$)			Mean D_{ik}
	Mean	S.D.	N	Mean	S.D.	N	All obs.
Flesch	9.39	7.80	52	−4.96	5.50	27	2.92 * ** (0.79)
Flesch Kincaid	1.65	1.40	63	−1.30	1.36	23	0.66 * ** (0.16)
Gunning Fog	2.31	1.85	55	−1.55	1.74	24	0.79 * ** (0.20)
SMOG	1.74	1.42	51	−1.00	1.12	27	0.51 * ** (0.15)
Dale-Chall	0.83	0.67	61	−0.67	0.50	25	0.29 * ** (0.08)
LLM Readability	0.77	0.58	72	−0.98	0.54	10	0.36 * ** (0.07)

Notes. Sample 121 matched pairs. First and second panels display conditional means, standard deviations and observation counts of D_{ik} (Corollary 1) from subpopulations of matched pairs in which the woman or man, respectively, satisfies Conditions 1 and 2. Male scores are subtracted from female scores, meaning D_{ik} is, by definition, positive in panel one and negative in panel two. Third panel averages D_{ik} over all matched pairs ($D_{ik} = 0$ whenever neither member simultaneously satisfies Conditions 1 and 2). Estimates are weighted by frequency observations are used in a match; degrees-of-freedom corrected standard errors in parentheses. ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

15 Table 9 LLM — D_{ik} (LLM tonal composites, base)

Table 14: D_{ik} (LLM tonal composites)

	Higher standards for women ($D_{ik} > 0$)			Higher standards for men ($D_{ik} < 0$)			Mean D_{ik}
	Mean	S.D.	N	Mean	S.D.	N	All obs.
G1: Creativity & Hedging	0.99	0.69	50	−0.93	0.72	32	0.16*
G2: Assertiveness & Voice	0.68	0.54	35	−0.89	0.70	49	(0.09) −0.13*
G3: Structural Directness	0.63	0.48	73	−0.73	0.36	16	(0.08) 0.29 * **
G4: Stance & Novelty	0.87	0.71	38	−0.88	0.60	43	(0.06) −0.03
G5: Support & Impact	1.55	1.10	40	−1.23	0.95	40	(0.08) 0.16 (0.13)

Notes. Sample matched pairs. First and second panels display conditional means, standard deviations and observation counts for matched pairs exhibiting higher standards for women and men, respectively. Uses LLM tonal composite scores. ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

16 Table 10

Table 15: Readability of authors' t th paper (draft and final)

	$t = 1$	$t = 2$	$t = 3$	$t = 4-5$	$t \geq 6$
Predicted $R_{jP} - R_{jW}$					
Women	2.09 * ** (0.72)	1.59 * * (0.71)	1.16 (0.86)	0.35 (1.12)	-0.17 (1.46)
Men	-0.20 (0.17)	-0.12 (0.10)	0.02 (0.09)	-0.21 (0.14)	-0.16 (0.17)
Marginal effect of female ratio					
Published article	1.67 (1.11)	2.17 * * (0.85)	2.67 * ** (0.86)	3.16 * ** (1.14)	3.66 * * (1.54)
Draft paper	-0.62 (1.31)	0.46 (0.94)	1.53 * * (0.75)	2.60 * ** (0.88)	3.68 * ** (1.23)
Diff.-in-diff.	2.29 * ** (0.76)	1.71 * * (0.76)	1.14 (0.93)	0.56 (1.20)	-0.02 (1.52)

Notes. Sample 4,289 observations. Panel one displays magnitude of predicted $R_{jP} - R_{jW}$ (the direct effect of peer review) for women and men over increasing t . Panel two estimates the marginal effect of an article's female ratio ($\beta_1 + \beta_2 \times t$), separately for draft papers and published articles. Figures from FGLS estimation of equation (15), weighted by N_{it} (see the data appendix). Control variables include citation count (asinh), max. T (author prominence) and max. t (author seniority), native speaker and editor and journal-year fixed effects. Standard errors clustered by editor and robust to cross-model correlation in parentheses. ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

17 Table 10 LLM

Table 16: LLM readability of authors' t th paper (draft and final)

	$t = 1$	$t = 2$	$t = 3$	$t = 4-5$	$t \geq 6$
Predicted $R_{jP} - R_{jW}$					
Women	0.42 * ** (0.10)	0.39 * ** (0.09)	0.36 * ** (0.09)	0.31 * ** (0.11)	0.33 * * (0.14)
Men	0.42 * ** (0.02)	0.43 * ** (0.01)	0.43 * ** (0.01)	0.41 * ** (0.01)	0.47 * ** (0.02)
Marginal effect of female ratio					
Published article	0.43 * ** (0.11)	0.44 * ** (0.08)	0.45 * ** (0.07)	0.46 * ** (0.08)	0.46 * ** (0.10)
Draft paper	0.44 * ** (0.14)	0.48 * ** (0.11)	0.52 * ** (0.10)	0.56 * ** (0.11)	0.60 * ** (0.14)
Diff.-in-diff.	-0.01 (0.11)	-0.04 (0.10)	-0.07 (0.10)	-0.10 (0.12)	-0.14 (0.14)

Notes. Sample 4,289 observations. Panel one displays magnitude of predicted $R_{jP} - R_{jW}$ (the direct effect of peer review) for women and men over increasing t . Panel two estimates the marginal effect of an article's female ratio ($\beta_1 + \beta_2 \times t$), separately for draft papers and published articles. Figures from FGLS estimation of equation (15), weighted by N_{it} (see the data appendix). Control variables include citation count (asinh), max. T (author prominence) and max. t (author seniority), native speaker and editor and journal-year fixed effects. Standard errors clustered by editor and robust to cross-model correlation in parentheses. ***, ** and * statistically significant at 1%, 5% and 10%, respectively.